

**CH-17 Mechanics &
Ch-18 Hydraulics
Homework Answers**



01

What is the horizontal force required to move a 250-lb wooden crate along a concrete floor? The surface area of the crate contacting the floor is 4 ft². The coefficient of friction between wood and a concrete floor is 0.62.

A. 38.75 lbs

$$F = \mu N$$

B. 38.75 N

C. **155 lbs**

$$F = \mu N = 0.62(250 \text{ lbs}) = 155 \text{ lbs}$$

D. 155 N



02

What is the kinetic energy of a 6 kg bowling ball rolling down the lane at 6 meters/sec?

A. 18 J

B. 36 J

C. 54 J

D. 108 J

$$\text{K.E.} = \frac{1}{2} mv^2$$

$$\text{KE} = \frac{1}{2} (6 \text{ kg}) (6 \text{ m/sec})^2 = 108 \text{ J}$$



03

You are asked to check the flow of a fire hydrant.

You get your pitot pressure gauge, stick the gauge

in the full flow from the fire hydrant and find that

the pitot pressure is 65 psi. What is the flow velocity in ft/sec

from this fire hydrant? Hint: use $P = 0.433 \times h$ and $v = \sqrt{2gh}$

A. 98 ft/sec

$$P = 0.433h$$

$$65 \text{ psi} = 0.433h$$

$$h = 65 / 0.433 = 150.12 \text{ ft of H}_2\text{O}$$

B. 102 ft/sec

C. 105 ft/sec

$$v = \sqrt{2(32.2 \text{ ft/sec}^2)(150.12 \text{ ft H}_2\text{O})} = 98.3 \text{ ft/sec}$$

D. 109 ft/sec

04

A two-piece block and tackle with 4 part of rope to lift a weight of 2,000 lbs. requires how much force for lift (no friction)?
_____ lbs.?

A. 300

B. 400

C. 500

D. 600

No. of falls = 4

Tension on each fall = $2000/4 = 500$



05

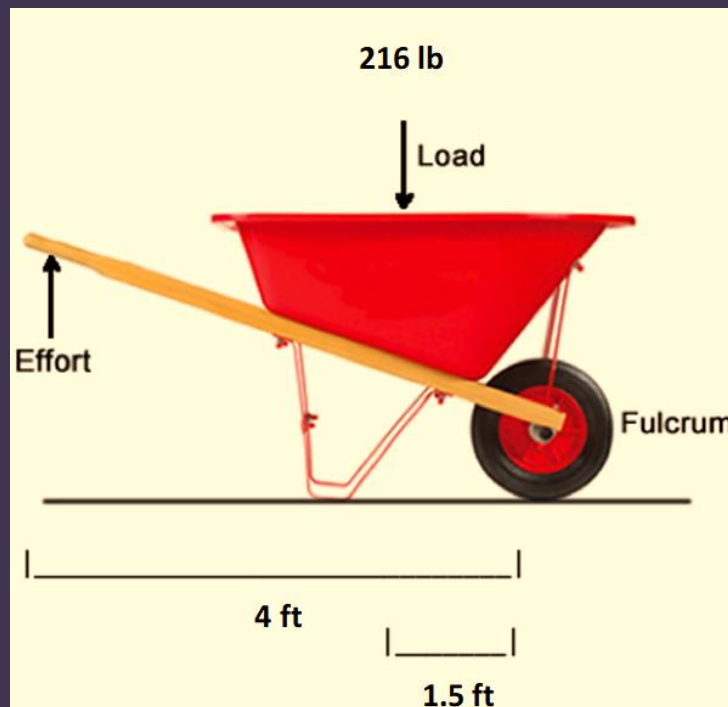
A wheelbarrow contains a load of 216 lb at an average distance of 1-1/2 ft from the axle. If the shaft of the wheelbarrow is 4 ft long, what force will be needed to raise the handles of the wheelbarrow?

A. 27 lb

B. 54 lb

C. 81 lb

D. 108 lb



$$216 \text{ lb} \times 1.5 \text{ ft} = 324 \text{ ft lb}$$

$$324 \text{ ft lb} = 4 \text{ ft} \times (f)$$

$$(f) = 324 / 4 = 81 \text{ lb}$$

06

An automobile traveling 55 mi/hr is slowing up at the rate of 5 mi/hr/sec.

How many seconds will it take to stop?

$$v = v_o + at$$

A. 6 sec

B. 0.09 sec

C. 11 sec

D. 0.40 sec

$$\text{Zero} = 55 \text{ mi/hr} + (-5 \text{ mi/hr/sec}) \times (t)$$

$$-55 \text{ mi/hr} = -5 \text{ mi/hr/sec} \times (t)$$

$$(t) = 55 \text{ mi/hr} \div 5 \text{ mi/hr/sec} = 11 \text{ sec}$$

07

An automatic sprinkler system, maintained at a system pressure of 60 psi, has an orifice discharge coefficient of 3.6. What is the delivery flow rate of the system?

$$P = \left(\frac{Q}{K} \right)^2$$

A. 27.9 gpm

B. 32.7 gpm

C. 36.2 gpm

D. 37.6 gpm

$$P=60\text{psi}$$

$$K=3.6$$

$$60 = \left(\frac{Q}{3.6} \right)^2$$

$$\sqrt{60} = \frac{Q}{3.6}$$

$$Q = 3.6\sqrt{60} = 27.9 \text{ gpm}$$



08

A large retail establishment is under construction. The contractor built the first wall before building the rest of the building. The first wall is 20 feet high and 300 feet long, and it is unsupported. A 15 mph wind is blowing broadside against the wall exerting a pressure of 0.011 psi. What is the total force against the wall?

$$P = \frac{F}{a}$$

A. 0.011 lbs

B. 95.0 lbs

C. 190 lbs

D. 9,500 lbs

Step 1: Calculate the area of the wall.

$$\text{Area} = 20 \text{ ft} \times 300 \text{ ft} = 6,000 \text{ ft}^2$$

Step 2: Convert the area to square inches.

$$6,000 \text{ ft}^2 \left(\frac{144 \text{ in}^2}{\text{ft}^2} \right) = 864,000 \text{ in}^2$$

Step 3: Calculate the force against the wall.

$$\text{Force} = 864,000 \text{ in}^2 \left(\frac{0.011 \text{ lbs}}{\text{in}^2} \right) = 9,500 \text{ lbs.}$$



09

If water weighs 8.34 lb/gal, what work is required to move 300 gallons of water from a 400 ft. deep mine?

$$W = Fs$$

A. 1,000,200 ft-lb

B. 1,000,400 ft-lb

C. 1,000,600 ft-lb

D. 1,000,800 ft-lb

$$W = (8.34 \text{ lb/gal} \times 300 \text{ gal}) \times 400 \text{ ft}$$

$$W = 1,000,800 \text{ ft-lb}$$



10

For 10 seconds, the velocity of a car which travels with a constant acceleration, changes from 10m/s to 30m/s. How far does the car travel?

$$v = v_o + at$$

$$v^2 = v_o^2 + 2as$$

A. 200m

$$(1) \quad 30 = 10 + a \times 10, \quad a = (30 - 10)/10 = 2$$

B. 300m

$$(2) \quad (30)^2 = (10)^2 + 2 \times 2 \times S$$

C. 400m

$$900 = 100 + 4S$$

D. 500m

$$S = (900 - 100)/4 = 800/4 = 200 \text{ meter}$$

11

The water level in a well is 46 ft (14.0 m) below the surface of the ground. A pump lifts the water to a holding tank where the outlet pipe is 21 ft (6.4 m) above the surface of the ground. Neglecting friction and other system losses, what is the theoretical head on the pump required for pumping water from the well to the holding tank?

- A. 21 ft (6.4 m)
- B. 25 ft (7.6 m)
- C. 46 ft (14.0 m)
- D. 67 ft (20.4 m)**

